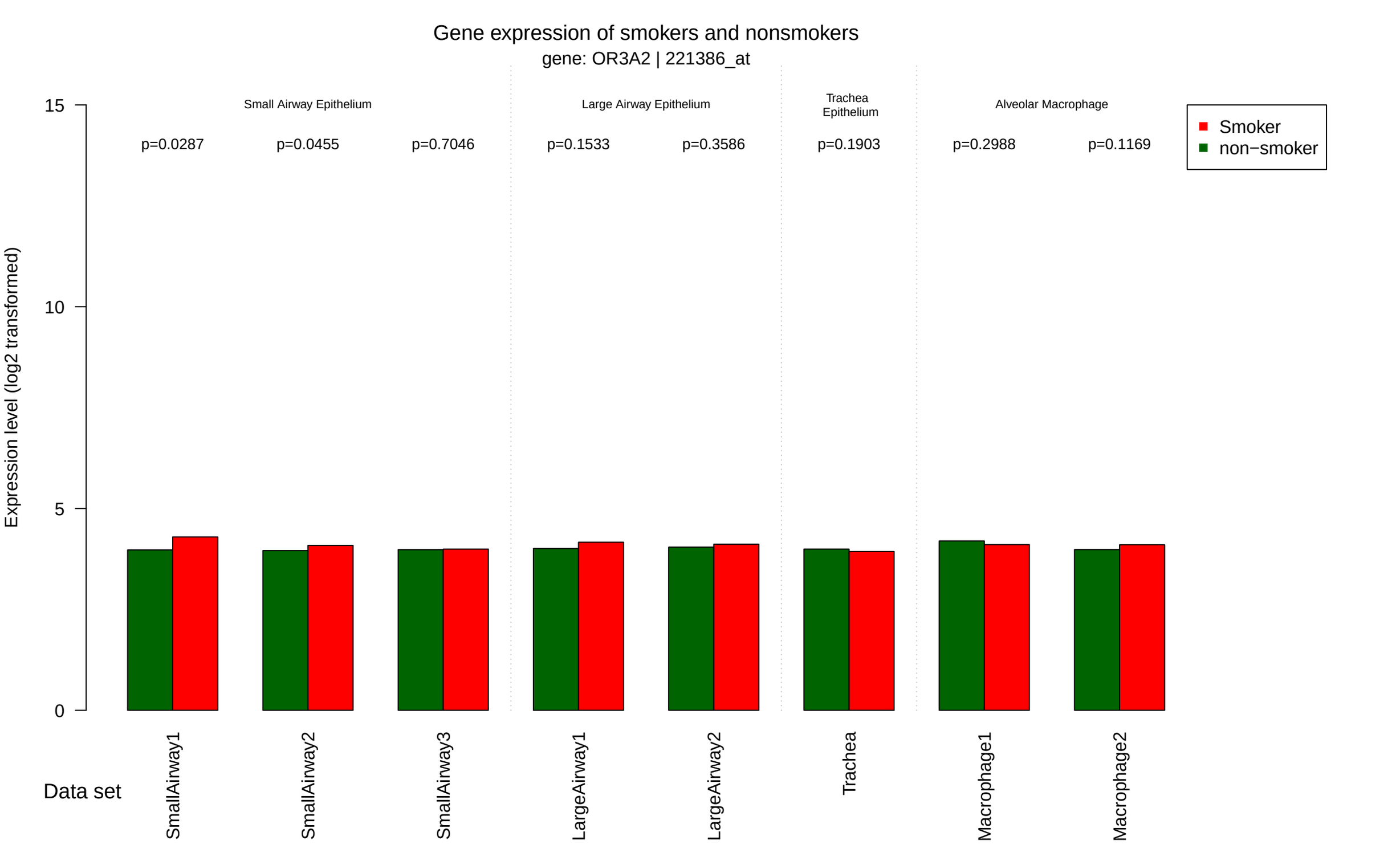
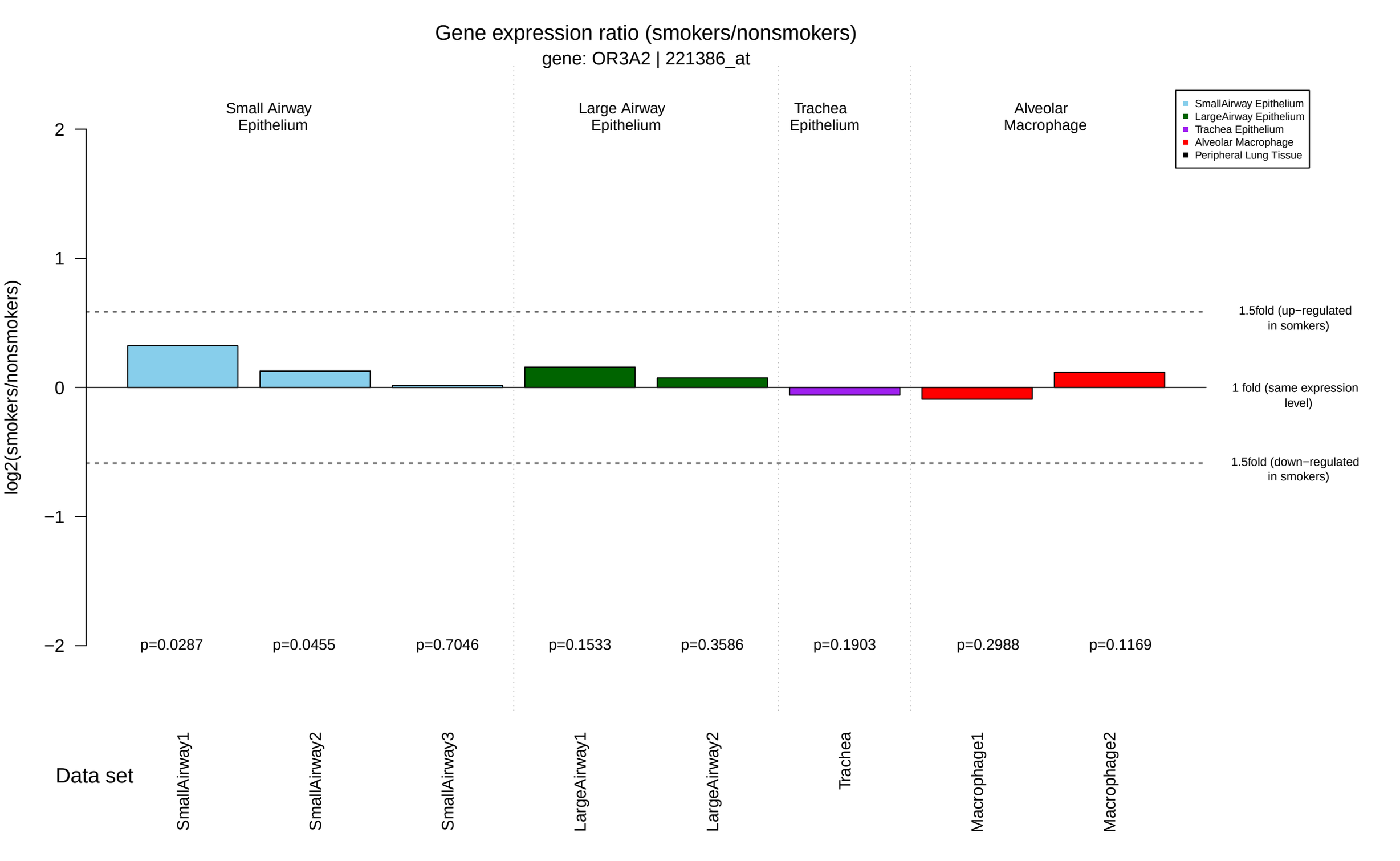
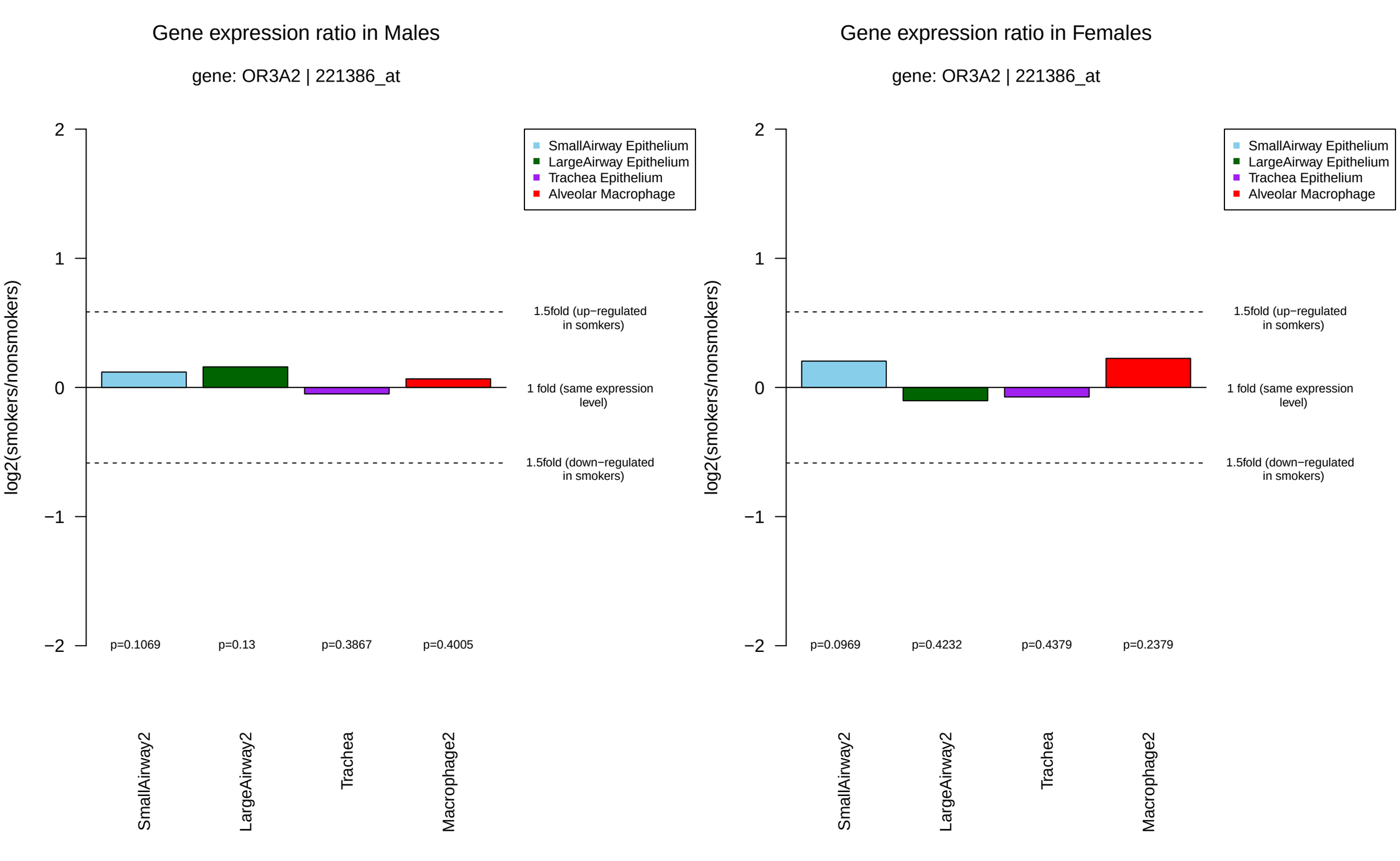
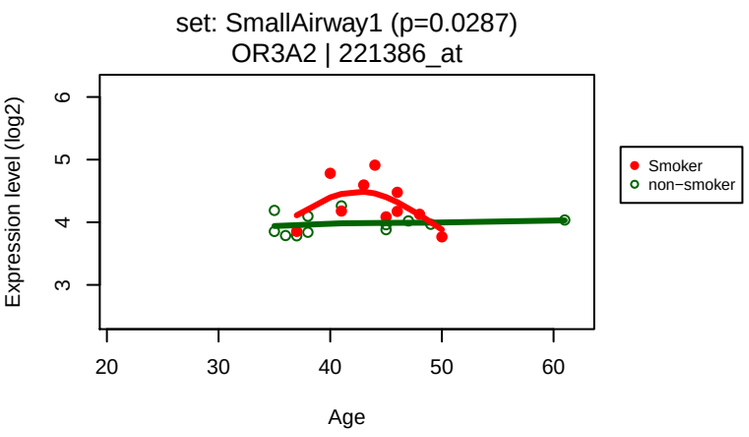
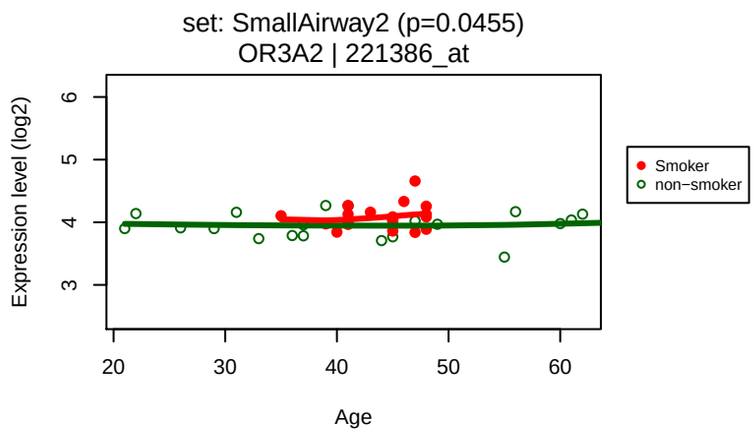
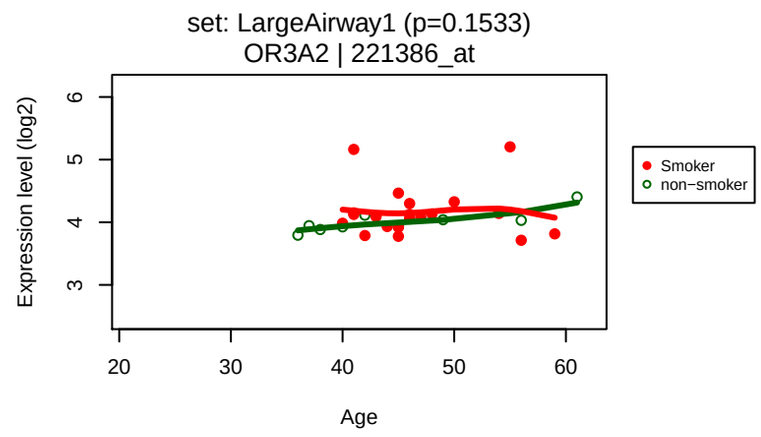

Gene expression of smokers and nonsmokers
gene: OR3A2 | 221386_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0287
p=0.0455
p=0.7046
p=0.1533
p=0.3586
p=0.1903
p=0.2988
p=0.1169
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

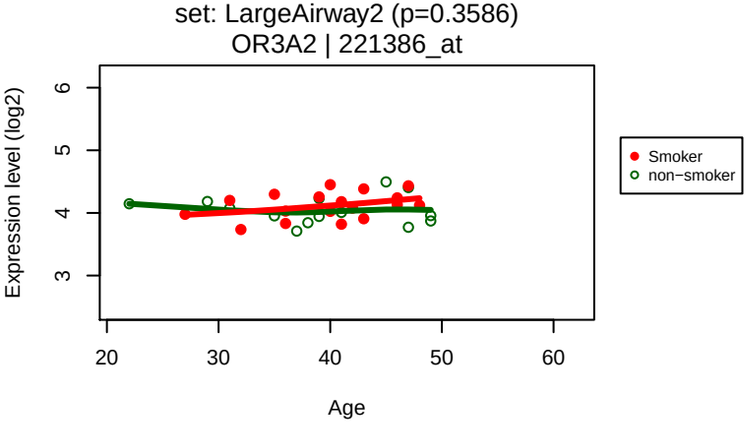

Gene expression ratio (smokers/nonsmokers)
gene: OR3A2 | 221386_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0287
p=0.0455
p=0.7046
p=0.1533
p=0.3586
p=0.1903
p=0.2988
p=0.1169
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

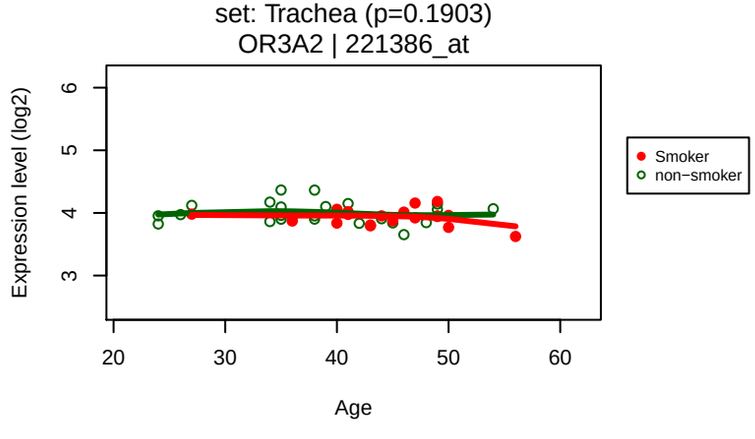

Gene expression ratio in Males
gene: OR3A2 | 221386_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1069
p=0.13
p=0.3867
p=0.4005
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR3A2 | 221386_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0969
p=0.4232
p=0.4379
p=0.2379
SmallAirway2
LargeAirway2
Trachea
Macrophage2

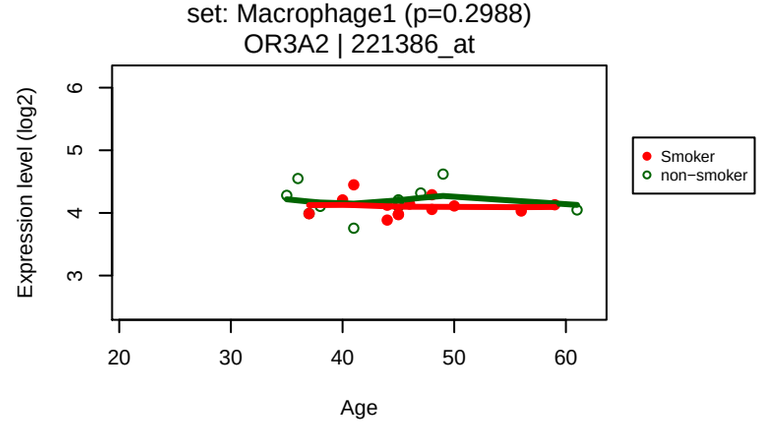

set: SmallAirway1 (p=0.0287)
OR3A2 | 221386_at
Expression level (log2)
Age
Smoker
non−smoker

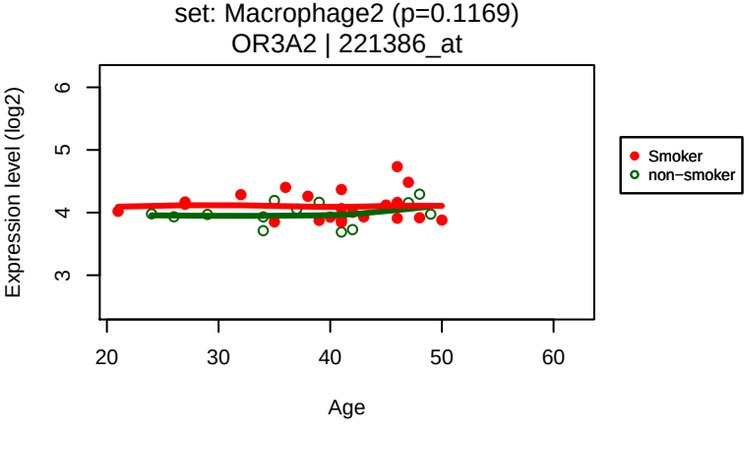

set: SmallAirway2 (p=0.0455)
OR3A2 | 221386_at
Expression level (log2)
Age
Smoker
non−smoker

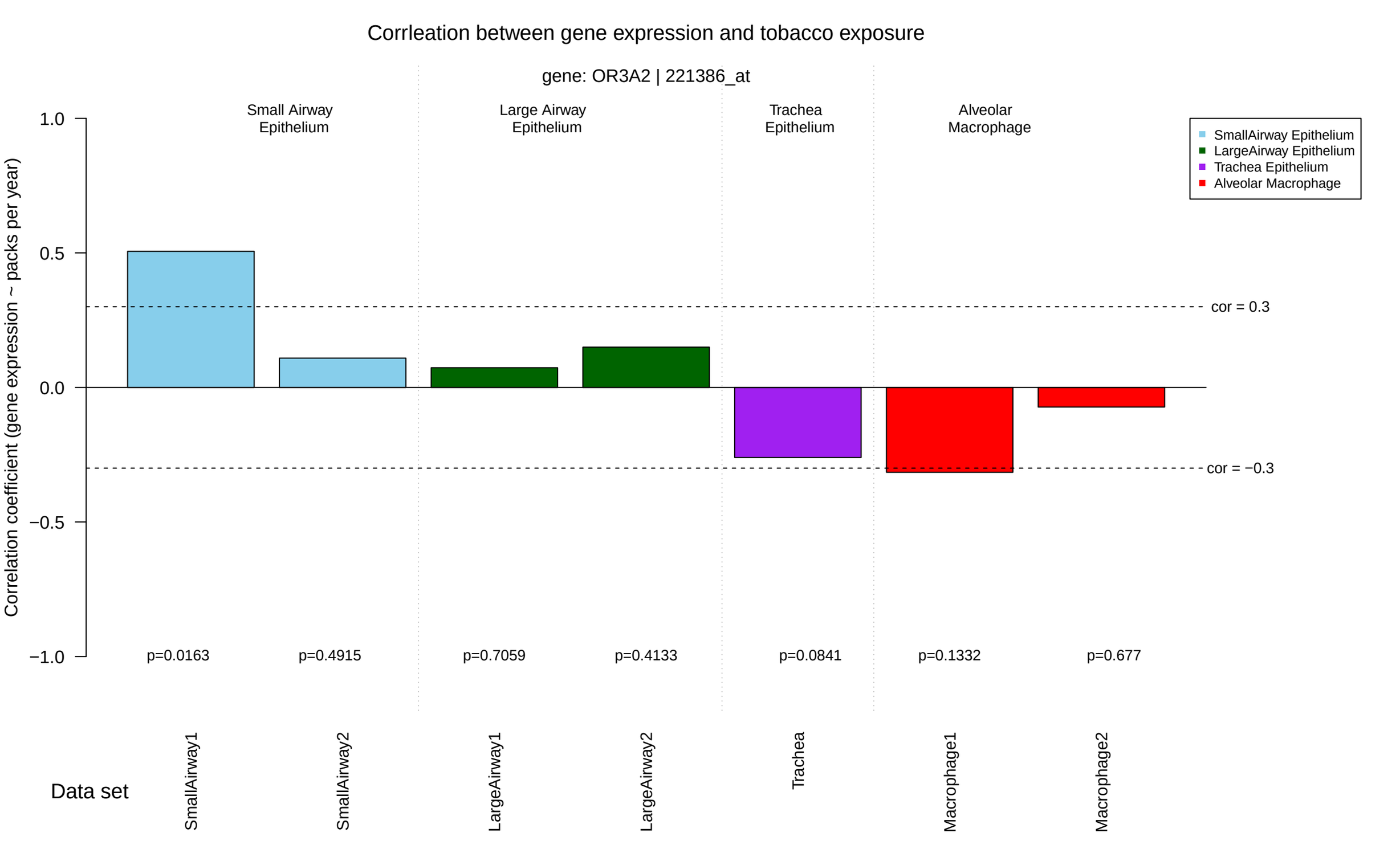

set: LargeAirway1 (p=0.1533)
OR3A2 | 221386_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3586)
OR3A2 | 221386_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1903)
OR3A2 | 221386_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2988)
OR3A2 | 221386_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1169)
OR3A2 | 221386_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR3A2 | 221386_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0163
p=0.4915
p=0.7059
p=0.4133
p=0.0841
p=0.1332
p=0.677
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2